

INMOTC: Geometry

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§0 Preliminaries

Problem 0.1. Prove the following exist in a triangle without using cyclic quadrilaterals:

- Circumcenter
- Incenter
- Orthocenter
- Centroid

Problem 0.2. If A, B, C, D are points in the plane such that no three are collinear then the following are equivalent:

- $ABCD$ is cyclic
- $\angle DAB = \angle DCB$
- $\angle ABC + \angle ADC = 180^\circ$

§1 Parallelograms

Problem 1.1. Let M and N be the midpoints of AB and AC in triangle ABC . Prove $2MN = BC$ without using similar triangles.

Problem 1.2. Let ABC be a triangle and M be the midpoint of BC . Squares $ABQP$ and $ACYX$ are erected. Show that $PX = 2AM$.

Problem 1.3. Let ABC be a triangle with circumcenter O and orthocenter H , and let M and N be the midpoints of AB and AC . Rays MO and NO meet line BC at Y and X , respectively. Lines MX and NY meet at P . Prove that OP bisects AH .

Problem 1.4. In a triangle ABC , if m_a, m_b, m_c are the medians. Let DEF be a triangle with these side lengths. Find the ratio of areas of ABC and DEF .

Problem 1.5 (Simson Line). Let ABC be a triangle with orthocenter H and let P be a point on the circumcircle of ABC . Let D, E, F be feet of perpendiculars from P to BC, AC, AB respectively. Prove that the midpoint of PH lies on the line DEF . (Prove that DEF are collinear. This is called the Simson Line.)

§2 Angle Chasing

Problem 2.1. In triangle ABC with circumcircle Γ , incenter I and A-excenter I_A , prove the following:

- If M_A is the midpoint of arc BC not containing A , $IM_A I_A$ are collinear.

- $IBI_A C$ are concyclic with center M_A .

Problem 2.2. Let H be the orthocenter of triangle ABC . Let the reflection of H across the BC be X and the reflection of H across the midpoint of BC be Y . Then X and Y both lie on the circumcircle of ABC . Moreover, AY is a diameter of the circumcircle.

Problem 2.3 (RMO 2016). Two circles C_1 and C_2 intersect each other at points A and B . Their external common tangent (closer to B) touches C_1 at P and C_2 at Q . Let C be the reflection of B in line PQ . Prove that $\angle CAP = \angle BAQ$.

Problem 2.4 (RMO 2019). In an acute angled triangle ABC , let H be the orthocenter, and let D, E, F be the feet of altitudes from A, B, C to the opposite sides, respectively. Let L, M, N be the midpoints of the segments AH, EF, BC respectively. Let X, Y be the feet of altitudes from L, N on to the line DF respectively. Prove that XM is perpendicular to MY .

Problem 2.5 (RMO 2019). Let ABC be a triangle with circumcircle Ω and let G be the centroid of triangle ABC . Extend AG, BG and CG to meet the circle Ω again in A_1, B_1 and C_1 . Suppose $\angle BAC = \angle A_1 B_1 C_1, \angle ABC = \angle A_1 C_1 B_1$ and $\angle ACB = \angle B_1 A_1 C_1$. Prove that ABC and $A_1 B_1 C_1$ are equilateral triangles.

Problem 2.6 (EGMO 2022/1). Let ABC be an acute-angled triangle in which $BC < AB$ and $BC < CA$. Let point P lie on segment AB and point Q lie on segment AC such that $P \neq B, Q \neq C$ and $BQ = BC = CP$. Let T be the circumcenter of triangle APQ , H the orthocenter of triangle ABC , and S the point of intersection of the lines BQ and CP . Prove that T, H , and S are collinear.

Problem 2.7 (INMO 2022). Let D be an interior point on the side BC of an acute-angled triangle ABC . Let the circumcircle of triangle ADB intersect AC again at $E (\neq A)$ and the circumcircle of triangle ADC intersect AB again at $F (\neq A)$. Let AD, BE , and CF intersect the circumcircle of triangle ABC again at $D_1 (\neq A), E_1 (\neq B)$ and $F_1 (\neq C)$, respectively. Let I and I_1 be the incentres of triangles DEF and $D_1 E_1 F_1$, respectively. Prove that E, F, I, I_1 are concyclic.

Problem 2.8 (ELMO 2021/1). In $\triangle ABC$, points P and Q lie on sides AB and AC , respectively, such that the circumcircle of $\triangle APQ$ is tangent to BC at D . Let E lie on side BC such that $BD = EC$. Line DP intersects the circumcircle of $\triangle CDQ$ again at X , and line DQ intersects the circumcircle of $\triangle BDP$ again at Y . Prove that D, E, X , and Y are concyclic.

Problem 2.9 (ISL 2021 G1). Let $ABCD$ be a parallelogram with $AC = BC$. A point P is chosen on the extension of ray AB past B . The circumcircle of ACD meets the segment PD again at Q . The circumcircle of triangle APQ meets the segment PC at R . Prove that lines CD, AQ, BR are concurrent.

Problem 2.10 (Steiner). Let ℓ be a line passing through orthocenter H of $\triangle ABC$. Let ℓ_a, ℓ_b, ℓ_c be the reflections of ℓ in BC, AC and AB respectively. Prove that ℓ_a, ℓ_b and ℓ_c concur on the circumcircle of triangle ABC .

§3 Power of Point and Radical Axis

Problem 3.1 (Power of Point(PoP)). Given a circle ω and point P . Consider any line AB with $A, B \in \omega$ and $P \in AB$. Let O be the center of ω and r be the circumradius of ω . Prove that $PA \cdot PB = PO^2 - r^2$. (This is called the power of P with respect to ω and is denoted by $\text{pow}_\omega(P)$.)

Problem 3.2 (Converse of PoP). Let $ABCD$ be a quadrilateral such that $P = AC \cap BD$ then $ABCD$ is concyclic iff $PA \cdot PC = PB \cdot PD$.

Problem 3.3 (Radical Axis). Let ω_1, ω_2 be two non-concentric circles. Prove that the locus of points P such that $pow_{\omega_1}(P) = pow_{\omega_2}(P)$ is a line perpendicular to the centers of ω_1 and ω_2 . (This is called the radical axis of ω_1 and ω_2 .)

Problem 3.4 (Radical Center). Let $\omega_1, \omega_2, \omega_3$ be three circles such that no two of them are concentric. Let ℓ_1 be the radical axis of ω_2 and ω_3 . Define ℓ_2 and ℓ_3 similarly. Prove that ℓ_1, ℓ_2 and ℓ_3 concur. (This point is called the radical center of ΔABC .)

Problem 3.5. Let ω_1, ω_2 be two intersecting circles with common tangent PQ and let AB be their common chord. Prove that if $P \in \omega_1$ and $Q \in \omega_2$ then AB bisects PQ .

Problem 3.6 (Euler's relation). If ΔABC has circumcenter, incenter, circumradius and inradius O, I, R, r respectively. Then prove that $OI^2 = R(R - 2r)$ and in particular $R \geq 2r$.

Problem 3.7 (INMO 2019). Let AB be the diameter of a circle Γ and let C be a point on Γ different from A and B . Let D be the foot of perpendicular from C on to AB . Let K be a point on the segment CD such that AC is equal to the semi perimeter of ADK . Show that the excircle of ADK opposite A is tangent to Γ .

Problem 3.8 (INMO 2020). Let Γ_1 and Γ_2 be two circles of unequal radii, with centres O_1 and O_2 respectively, intersecting in two distinct points A and B . Assume that the centre of each circle is outside the other circle. The tangent to Γ_1 at B intersects Γ_2 again in C , different from B ; the tangent to Γ_2 at B intersects Γ_1 again at D , different from B . The bisectors of $\angle DAB$ and $\angle CAB$ meet Γ_1 and Γ_2 again in X and Y , respectively. Let P and Q be the circumcentres of triangles ACD and XAY , respectively. Prove that PQ is the perpendicular bisector of the line segment O_1O_2 .

Problem 3.9 (IMO 2008). Let H be the orthocenter of an acute-angled triangle ABC . The circle Γ_A centered at the midpoint of BC and passing through H intersects the sideline BC at points A_1 and A_2 . Similarly, define the points B_1, B_2, C_1 and C_2 . Prove that the six points A_1, A_2, B_1, B_2, C_1 and C_2 are concyclic.

Problem 3.10 (ISL 2011). Let $ABCD$ be a convex quadrilateral whose sides AD and BC are not parallel. Suppose that the circles with diameters AB and CD meet at points E and F inside the quadrilateral. Let ω_E be the circle through the feet of the perpendiculars from E to the lines AB, BC and CD . Let ω_F be the circle through the feet of the perpendiculars from F to the lines CD, DA and AB . Prove that the midpoint of the segment EF lies on the line through the two intersections of ω_E and ω_F .

§4 Constructions

Problem 4.1 (Standard Constructions). Construct the following using straightedge and compass:

- The perpendicular through a given point to a given line.
- The line parallel to a given line through a given point
- The perpendicular bisector of a given segment
- The circumcenter, incenter, orthocenter and centroid of a given triangle.
- Divide a given line into a given rational ratio $\frac{p}{q}$.

Problem 4.2 (RMO 2017). Let AOB be a given angle less than 180° and let P be an interior point of the angular region determined by $\angle AOB$. Show, with proof, how to construct, using only ruler and compass, a line segment CD passing through P such that C lies on the ray OA and D lies on the ray OB , and $CP : PD = 1 : 2$.

Problem 4.3 (INMO 2021). Betal marks 2021 points on the plane such that no three are collinear, and draws all possible segments joining these. He then chooses any 1011 of these segments, and marks their midpoints. Finally, he chooses a segment whose midpoint is not marked yet, and challenges Vikram to construct its midpoint using only a straightedge. Can Vikram always complete this challenge?

Note. A straightedge is an infinitely long ruler without markings, which can only be used to draw the line joining any two given distinct points.

Problem 4.4 (Cute Sloth). Given two points B and C , and a straightedge and a pencil, extend the line BC in both directions without disturbing a point sized sloth sleeping on the line BC . (you can leave out a small segment around the sloth of length at most 1 unit.)

Problem 4.5 (Extended ELMO 2021). Janabel has a device that, when given two distinct points U and V in the plane, draws the perpendicular bisector of UV . Prove the following:

- Suppose X, Y lie on a drawn line ℓ . Suppose lines m, n are perpendicular to X, Y and are drawn. Then we can draw the reflection of X across Y .
- Suppose $S_0, S_1, \dots, S_5$ are equally spaced points in that order on a line ℓ . We can draw in the reflection of S_0 across S_4 , if S_0, S_4 and ℓ are marked.
- Given a point X on a line ℓ , we can draw the perpendicular to ℓ through X .
- Given a point X and a line ℓ , we can draw the perpendicular to ℓ through X . We can also draw the line parallel to ℓ through X .
- If A is a point on line l and B is an arbitrary point, we can draw AB given A, B and l .
- Janabel's tool is superior to the straightedge.
- Given three points A, B, C with no two collinear. You can draw the circumcenter, orthocenter and centroid of $\triangle ABC$.
- It is not necessary that we can draw the incenter of given $\triangle ABC$.

§5 Plane Transformations and Homothety

A homothety is a transformation of the plane by taking a point P and taking the map Φ_r such that $\Phi_r(Q)$, Q and P are collinear and $\frac{P\Phi_r(Q)}{PQ} = r$.

Problem 5.1. A composition of homotheties from different points is a homothety.

Problem 5.2. If we have a transformation Φ of the plane such that $\angle\Phi(P)\Phi(Q)\Phi(R) = \angle PQR$ for any points P, Q, R in the plane. Then Φ is a composition of a homothety and a rotation and a translation.

Problem 5.3 (Monge's theorem). If ω_1, ω_2 and ω_3 are three circles such that none of them are inside one another. Let P, Q and R be the intersection of their pairwise direct common tangents. Prove that P, Q and R are collinear.

Problem 5.4 (Homothetic Triangles). If ABC and XYZ are triangles then they are homothetic iff $AB \parallel XY$, $AC \parallel XZ$ and $BC \parallel YZ$.

Problem 5.5 (Nine Point Circle). In triangle ABC with circumcenter H , let X, Y and Z be the midpoints of AH, BH and CH respectively. Let L, M and N be the midpoints of AB, BC and AC respectively. Let D, E and F be the feet of A, B and C altitudes onto the opposite sides. Prove that $DEFXYZLMN$ all lie on a circle with radius half of ABC 's circumradius.

Problem 5.6 (Euler). In triangle ABC , prove that circumcenter O , centroid G and orthocenter H and find $\frac{OG}{GH}$.

§6 Miquel Point and Spiral Similarity

Problem 6.1 (Triangle Miquel). If in $\triangle ABC$, we have points P, Q and R on sides BC, AC and AB respectively then $(AQR), (BRP)$ and (CPR) concur.

A spiral similarity is the composition of a homothety and rotation from a single point.

Problem 6.2. For every quadruple of points A, B, C, D , there exists a unique spiral similarity that creates the two maps- $A \mapsto C, B \mapsto D$ unless $ABDC$ is a parallelogram.

Problem 6.3 (Spiral Similarities come in pairs). The the center of spiral similarity of $AB \mapsto CD$ is also the spiral center that takes $AC \mapsto BD$.

Problem 6.4 (Quadrilateral Miquel). Given a complete quadrilateral¹ $ABCDEF$, the circles $(EAB), (EBC), (FAB), (FCD)$ concur at a point M known as the Miquel Point M . M is now the center of six spiral similarities or three pairs of spiral similarities.

- $\overline{AB} \leftrightarrow \overline{CD}$ and $\overline{AD} \leftrightarrow \overline{BC}$
- $\overline{FD} \leftrightarrow \overline{DC}$ and $\overline{FB} \leftrightarrow \overline{DE}$
- $\overline{FA} \leftrightarrow \overline{CE}$ and $\overline{FC} \leftrightarrow \overline{AE}$

Problem 6.5. The feet of altitude from M to the 4 sides of the complete quadrilateral are collinear.

Problem 6.6. The circles with diameter AC, BD, EF are coaxial and the orthocenters of EBC, EAD, FAB, FCD lie on their radical axis. Their radical axis is known as the Aubert Line and this line is perpendicular to the line in 6.5.

Problem 6.7. The centers of $(EBC), (EAD), (FAB), (FCD)$ and M are concyclic.

Problem 6.8 (Cyclic Miquel). Assume that the quadrilateral $ABCD$ is cyclic with center O circumcircle ω and diagonals AC, BD intersecting at K . Prove the following:

- M is the inverse of K with respect to ω .
- M lies on $\overline{EF}$ and $OM \perp EF$.
- B, M, O, D are concyclic.
- C, M, O, A are concyclic.
- MO bisects $\angle BMD$ and $\angle AMC$.

Problem 6.9 (Brazil 2011/5). Let ABC be an acute triangle and H is orthocenter. Let D be the intersection of BH and AC and E be the intersection of CH and AB . The circumcircle of ADE cuts the circumcircle of ABC at $F \neq A$. Prove that the angle bisectors of $\angle BFC$ and $\angle BHC$ concur at a point on BC .

¹Given a quadrilateral $ABCD$ (possibly convex), let $E = AB \cap CD, F = AD \cap BC$. $ABCDEF$ is now called a complete quadrilateral. Thus, a complete quadrilateral $ABCDEF$ is determined by 4 lines and their 6 unique intersections. Imagine it as all 6 intersections of 4 given lines.

Problem 6.10 (USAMO 2006). Let $ABCD$ be a quadrilateral, and let E and F be points on sides AD and BC , respectively, such that $\frac{AE}{ED} = \frac{BF}{FC}$. Ray FE meets rays BA and CD at S and T , respectively. Prove that the circumcircles of triangles SAE , SBF , TCF , and TDE pass through a common point.

Problem 6.11 (USEMO 2019/1). Let $ABCD$ be a cyclic quadrilateral. A circle centered at O passes through B and D and meets lines BA and BC again at points E and F (distinct from A, B, C). Let H denote the orthocenter of triangle DEF . Prove that if lines AC , DO , EF are concurrent, then triangles ABC and EHF are similar.

Problem 6.12 (ISL 2015 G3). Let ABC be a triangle with $\angle C = 90^\circ$, and let H be the foot of the altitude from C . A point D is chosen inside the triangle CBH so that CH bisects AD . Let P be the intersection point of the lines BD and CH . Let ω be the semicircle with diameter BD that meets the segment CB at an interior point. A line through P is tangent to ω at Q . Prove that the lines CQ and AD meet on ω .

Problem 6.13 (USATSTST 2012/2). Let $ABCD$ be a quadrilateral with $AC = BD$. Diagonals AC and BD meet at P . Let ω_1 and O_1 denote the circumcircle and the circumcenter of triangle ABP . Let ω_2 and O_2 denote the circumcircle and circumcenter of triangle CDP . Segment BC meets ω_1 and ω_2 again at S and T (other than B and C), respectively. Let M and N be the midpoints of minor arcs $\widehat{SP}$ (not including B) and $\widehat{TP}$ (not including C). Prove that $MN \parallel O_1O_2$.

§7 Combined

Problem 7.1 (INMO 2021). In a convex quadrilateral $ABCD$, $\angle ABD = 30^\circ$, $\angle BCA = 75^\circ$, $\angle ACD = 25^\circ$ and $CD = CB$. Extend CB to meet the circumcircle of triangle DAC at E . Prove that $CE = BD$.

Problem 7.2 (LMAO 2021). Let I, O and Γ respectively be the incentre, circumcentre and circumcircle of triangle ABC . Points A_1, A_2 are chosen on Γ , such that $AA_1 = AI = AA_2$, and point A' is the foot of the altitude from I to A_1A_2 . If B', C' are similarly defined, prove that lines AA', BB' and CC' concur on OI .

Problem 7.3 (ISL 2011). Let ABC be a triangle with $AB = AC$ and let D be the midpoint of AC . The angle bisector of $\angle BAC$ intersects the circle through D, B and C at the point E inside the triangle ABC . The line BD intersects the circle through A, E and B in two points B and F . The lines AF and BE meet at a point I , and the lines CI and BD meet at a point K . Show that I is the incentre of triangle KAB .

Problem 7.4 (EGMO TST 2022). Let ABC be an isosceles triangle with $AB = AC$. Suppose P, Q, R are points on segments AC, AB, BC respectively such that $AP = QB$, $\angle PBC = 90^\circ - \angle BAC$ and $RP = RQ$. Let O_1, O_2 be the circumcenters of $\triangle APQ$ and $\triangle CRP$. Prove that $BR = O_1O_2$.

Problem 7.5 (IGO 2021). Acute-angled triangle ABC with circumcircle ω is given. Let D be the midpoint of AC , E be the foot of altitude from A to BC , and F be the intersection point of AB and DE . Point H lies on the arc BC of ω (the one that does not contain A) such that $\angle BHE = \angle ABC$. Prove that $\angle BHF = 90^\circ$.

Problem 7.6 (Poland). Let X be a point lying in the interior of the acute triangle ABC such that

$$\sphericalangle BAX = 2\sphericalangle XBA \quad \text{and} \quad \sphericalangle XAC = 2\sphericalangle ACX.$$

Denote by M the midpoint of the arc BC of the circumcircle (ABC) containing A . Prove that $XM = XA$.

Problem 7.7 (RMM 2019/2). Let $ABCD$ be an isosceles trapezoid with $AB \parallel CD$. Let E be the midpoint of AC . Denote by ω and Ω the circumcircles of the triangles ABE and CDE , respectively. Let P be the crossing point of the tangent to ω at A with the tangent to Ω at D . Prove that PE is tangent to Ω .

Problem 7.8 (IMO 2018/1). Let Γ be the circumcircle of acute triangle ABC . Points D and E are on segments AB and AC respectively such that $AD = AE$. The perpendicular bisectors of BD and CE intersect minor arcs AB and AC of Γ at points F and G respectively. Prove that lines DE and FG are either parallel or they are the same line.

Problem 7.9 (IMO 2019/2). In triangle ABC , point A_1 lies on side BC and point B_1 lies on side AC . Let P and Q be points on segments AA_1 and BB_1 , respectively, such that PQ is parallel to AB . Let P_1 be a point on line PB_1 , such that B_1 lies strictly between P and P_1 , and $\angle PP_1C = \angle BAC$. Similarly, let Q_1 be the point on line QA_1 , such that A_1 lies strictly between Q and Q_1 , and $\angle CQ_1Q = \angle CBA$. Prove that points P, Q, P_1 , and Q_1 are concyclic.

Problem 7.10 (ISL 2019). Let I be the incentre of acute-angled triangle ABC . Let the incircle meet BC, CA , and AB at D, E , and F , respectively. Let line EF intersect the circumcircle of the triangle at P and Q , such that F lies between E and P . Prove that $\angle DPA + \angle AQD = \angle QIP$.

Problem 7.11 (ELMOSL 2012). ABC is a triangle with incenter I . The foot of the perpendicular from I to BC is D , and the foot of the perpendicular from I to AD is P . Prove that $\angle BPD = \angle DPC$.

Problem 7.12 (USEMO 2020). The sides of a convex 200-gon $A_1A_2 \dots A_{200}$ are colored red and blue in an alternating fashion. Suppose the extensions of the red sides determine a regular 100-gon, as do the extensions of the blue sides. Prove that the 50 diagonals $\overline{A_1A_{101}}, \overline{A_3A_{103}}, \dots, \overline{A_{99}A_{199}}$ are concurrent.

Problem 7.13 (ISL 2011). Let ABC be a triangle with incenter I and circumcircle ω . Let D and E be the second intersection points of ω with AI and BI , respectively. The chord DE meets AC at a point F , and BC at a point G . Let P be the intersection point of the line through F parallel to AD and the line through G parallel to BE . Suppose that the tangents to ω at A and B meet at a point K . Prove that the three lines AE, BD and KP are either parallel or concurrent.

Problem 7.14 (Russia 2021). In triangle ABC angle bisectors AA_1 and CC_1 intersect at I . Line through B parallel to AC intersects rays AA_1 and CC_1 at points A_2 and C_2 respectively. Let O_a and O_c be the circumcenters of triangles AC_1C_2 and CA_1A_2 respectively. Prove that $\angle O_aBO_c = \angle AIC$.

Problem 7.15 (Russia 2011). Let N be the midpoint of arc ABC of the circumcircle of triangle ABC , let M be the midpoint of AC and let I_1, I_2 be the incentres of triangles ABM and CBM . Prove that points I_1, I_2, B, N lie on a circle.

Problem 7.16 (LMAO 2021). Let P, Q be chosen inside $\triangle ABC$ such that $\angle PBC = \angle ABQ$ and $\angle PCB = \angle ACQ$. The tangent to (BPC) at P and the tangent to (BQC) at Q intersect at R . Prove that circles (PQR) and (ABC) , and line AR share a common point.

Problem 7.17 (IMO 2011). Let ABC be an acute triangle with circumcircle Γ . Let ℓ be a tangent line to Γ , and let ℓ_a, ℓ_b and ℓ_c be the lines obtained by reflecting ℓ in the lines BC, CA and AB , respectively. Show that the circumcircle of the triangle determined by the lines ℓ_a, ℓ_b and ℓ_c is tangent to the circle Γ .

Remark. Bonus sections 1 and 2 are completely irrelevant for INMO and are kept here only for fun and those who are aware what these techniques are can use them as some fun practice. All the best!

§8 Bonus 1: Inversion

Problem 8.1 (ELMO 2018/3). Let A be a point in the plane, and ℓ a line not passing through A . Evan does not have a straightedge, but instead has a special compass which has the ability to draw a circle through three distinct noncollinear points. (The center of the circle is not marked in this process.) Additionally, Evan can mark the intersections between two objects drawn, and can mark an arbitrary point on a given object or on the plane.

(i) Can Evan construct* the reflection of A over ℓ ?

(ii) Can Evan construct the foot of the altitude from A to ℓ ?

*To construct a point, Evan must have an algorithm which marks the point in finitely many steps.

Problem 8.2 (IGO). Two circles Γ_1 and Γ_2 meet at two distinct points A and B . A line passing through A meets Γ_1 and Γ_2 again at C and D respectively, such that A lies between C and D . The tangent at A to Γ_2 meets Γ_1 again at E . Let F be a point on Γ_2 such that F and A lie on different sides of BD , and $2\angle AFC = \angle ABC$. Prove that the tangent at F to Γ_2 , and lines BD and CE are concurrent.

Problem 8.3 (IGO). Consider a triangle ABC with altitudes AD, BE , and CF , and orthocenter H . Let the perpendicular line from H to EF intersect EF, AB and AC at P, T and L , respectively. Point K lies on the side BC such that $BD = KC$. Let ω be a circle that passes through H and P , that is tangent to AH . Prove that circumcircle of triangle ATL and ω are tangent, and KH passes through the tangency point.

Problem 8.4 (IMO 2015/3). Let ABC be an acute triangle with $AB > AC$. Let Γ be its circumcircle, H its orthocenter, and F the foot of the altitude from A . Let M be the midpoint of BC . Let Q be the point on Γ such that $\angle HQA = 90^\circ$ and let K be the point on Γ such that $\angle HKQ = 90^\circ$. Assume that the points A, B, C, K and Q are all different and lie on Γ in this order. Prove that the circumcircles of triangles KQH and FKM are tangent to each other.

Problem 8.5 (RMM 2018/6). Fix a circle Γ , a line ℓ tangent to Γ , and another circle Ω disjoint from ℓ such that Γ and Ω lie on opposite sides of ℓ . The tangents to Γ from a variable point X on Ω meet ℓ at Y and Z . Prove that, as X varies over Ω , the circumcircle of XYZ is tangent to two fixed circles.

Problem 8.6 (Russia 2013/11.8). Let ω be the incircle of the triangle ABC and with centre I . Let Γ be the circumcircle of the triangle AIB . Circles ω and Γ intersect at the point X and Y . Let Z be the intersection of the common tangents of the circles ω and Γ . Show that the circumcircle of the triangle XYZ is tangent to the circumcircle of the triangle ABC .

Problem 8.7 (IMO 2021/3). Let D be an interior point of the acute triangle ABC with $AB > AC$ so that $\angle DAB = \angle CAD$. The point E on the segment AC satisfies $\angle ADE = \angle BCD$, the point F on the segment AB satisfies $\angle FDA = \angle DBC$, and the point X on the line AC satisfies $CX = BX$. Let O_1 and O_2 be the circumcenters of the triangles ADC and EXD , respectively. Prove that the lines BC , EF , and O_1O_2 are concurrent.

Problem 8.8 (IMO 2018/6). A convex quadrilateral $ABCD$ satisfies $AB \cdot CD = BC \cdot DA$. Point X lies inside $ABCD$ so that

$$\angle XAB = \angle XCD \quad \text{and} \quad \angle XBC = \angle XDA.$$

Prove that $\angle BXA + \angle DXC = 180^\circ$.

§9 Bonus 2: Projective

Problem 9.1 (AoPS). Let $PQRS$ be an isosceles trapezium with $PQ \parallel RS$, and Ω be its circumcircle. Let X be a point on the arc PQ or RS of Ω , and Y be a point on the segment PQ . Also, Let $XY \cap RS = A$, $PA \cap SX = B$, $QA \cap RX = C$. Then Y is the midpoint of segment PQ if and only if XY , BR , CS are concurrent.

Problem 9.2 (ISL 2019). Let P be a point inside triangle ABC . Let AP meet BC at A_1 , let BP meet CA at B_1 , and let CP meet AB at C_1 . Let A_2 be the point such that A_1 is the midpoint of PA_2 , let B_2 be the point such that B_1 is the midpoint of PB_2 , and let C_2 be the point such that C_1 is the midpoint of PC_2 . Prove that points A_2 , B_2 , and C_2 cannot all lie strictly inside the circumcircle of triangle ABC .

Problem 9.3 (Micheal Villiers). Let $ABCDEF$ be a hexagon with opposite sides parallel. Prove that the three lines joining the midpoints of opposites sides are concurrent.

Problem 9.4 (USA January TST 2015/3). Let ABC be a non-equilateral triangle and let M_a , M_b , M_c be the midpoints of the sides BC , CA , AB , respectively. Let S be a point lying on the Euler line. Denote by X , Y , Z the second intersections of M_aS , M_bS , M_cS with the nine-point circle. Prove that AX , BY , CZ are concurrent.

Problem 9.5 (RMM 2012/6). Let ABC be a triangle and let I and O denote its incentre and circumcentre respectively. Let ω_A be the circle through B and C which is tangent to the incircle of the triangle ABC ; the circles ω_B and ω_C are defined similarly. The circles ω_B and ω_C meet at a point A' distinct from A ; the points B' and C' are defined similarly. Prove that the lines AA' , BB' and CC' are concurrent at a point on the line IO .

Problem 9.6 (ISL 2004 G8). Given a cyclic quadrilateral $ABCD$, let M be the midpoint of the side CD , and let N be a point on the circumcircle of triangle ABM . Assume that the point N is different from the point M and satisfies $\frac{AN}{BN} = \frac{AM}{BM}$. Prove that the points E , F , N are collinear, where $E = AC \cap BD$ and $F = BC \cap DA$.

Problem 9.7 (RMM 2013/3). Let $ABCD$ be a quadrilateral inscribed in a circle ω . The lines AB and CD meet at P , the lines AD and BC meet at Q , and the diagonals AC and BD meet at R . Let M be the midpoint of the segment PQ , and let K be the common point of the segment MR and the circle ω . Prove that the circumcircle of the triangle KPQ and ω are tangent to one another.

Problem 9.8 (Taiwan TST Quiz 3/3/2 2015). In a scalene triangle ABC with incenter I , the incircle is tangent to sides CA and AB at points E and F . The tangents to the circumcircle of triangle AEF at E and F meet at S . Lines EF and BC intersect at T . Prove that the circle with diameter ST is orthogonal to the nine-point circle of triangle BIC .

Problem 9.9 (USATSTST 2016/6). Let ABC be a triangle with incenter I , and whose incircle is tangent to $\overline{BC}$, $\overline{CA}$, $\overline{AB}$ at D , E , F , respectively. Let K be the foot of the altitude from D to $\overline{EF}$. Suppose that the circumcircle of $\triangle AIB$ meets the incircle at two distinct points C_1 and C_2 , while the circumcircle of $\triangle AIC$ meets the incircle at two distinct points B_1 and B_2 . Prove that the radical axis of the circumcircles of $\triangle BB_1B_2$ and $\triangle CC_1C_2$ passes through the midpoint M of $\overline{DK}$.

Problem 9.10 (ISL 2012/G8). Let ABC be a triangle with circumcircle ω and ℓ a line without common points with ω . Denote by P the foot of the perpendicular from the center of ω to ℓ . The side-lines BC, CA, AB intersect ℓ at the points X, Y, Z different from P . Prove that the circumcircles of the triangles AXP , BYP and CZP have a common point different from P or are mutually tangent at P .

§10 Bonus 3: Combinatorial Geometry

This section is purely bonus because some people² do not like Combo Geo as Geo. Do enjoy!

Problem 10.1 (EGMO TST 2022). Let $r > 0$ be a real number. All the interior points of the disc $D(r)$ of radius r are colored with one of two colors, red or blue.

- If $r > \frac{\pi}{\sqrt{3}}$, show that we can find two points A and B in the interior of the disc such that $AB = \pi$ and A, B have the same color.
- Does the conclusion in (a) hold if $r > \frac{\pi}{2}$?

Problem 10.2 (Folklore). Given $2n$ points in a plane with no three collinear, with n red points and n blue points, prove that there exists a pairing of the red and blue points such that then segments joining each pair are pairwise non-intersecting.

Problem 10.3 (Folklore). Finitely many lines divide the plane into regions. Show that these regions can be colored by two colors in such a way that neighboring regions have different colors.

Problem 10.4 (Unknown). An equilateral triangle T and a finite set of points in the plane S are drawn, so that any choice of 9 points of S can be covered by some translate of T . Prove that the entire set S can be covered using 2 translates of T .

Problem 10.5 (Unknown). Some congruent paper squares of k distinct colors are placed on a rectangular table, with sides parallel to the sides of the table. Suppose that for any k squares of distinct colors, some two of them can be nailed on the table with only one nail. Prove that there is a color such that all squares of that color can be nailed with $2k - 2$ nails.

Problem 10.6 (Folklore). A set of points is marked on the plane, with the property that any three marked points can be covered with a disk of radius 1. Prove that the set of all marked points can be covered with a disk of radius 1.

²with questionable taste

Problem 10.7. We tile a convex polygon with 2021 parallelograms. Prove that we could have tiled the same convex polygon with 2020 parallelograms.

Problem 10.8 (Korea 2015). There are 2015 distinct circles in a plane, with radius 1. Prove that you can select 27 circles, which form a set C , which satisfy the following:

- For two arbitrary circles in C , they intersect with each other or
- For two arbitrary circles in C , they don't intersect with each other.

Problem 10.9 (Sylvester-Gallai). Given a finite set S of points in a plane, either they all lie on a line or there is a line containing exactly two points of S .

Problem 10.10 (ISL 2020). In the plane, there are $n \geq 6$ pairwise disjoint disks $D_1, D_2, \dots, D_n$ with radii $R_1 \geq R_2 \geq \dots \geq R_n$. For every $i = 1, 2, \dots, n$, a point P_i is chosen in disk D_i . Let O be an arbitrary point in the plane. Prove that

$$OP_1 + OP_2 + \dots + OP_n \geq R_6 + R_7 + \dots + R_n.$$

Problem 10.11 (ISL 2019). On a flat plane in Camelot, King Arthur builds a labyrinth $\mathcal{L}$ consisting of n walls, each of which is an infinite straight line. No two walls are parallel, and no three walls have a common point. Merlin then paints one side of each wall entirely red and the other side entirely blue.

At the intersection of two walls there are four corners: two diagonally opposite corners where a red side and a blue side meet, one corner where two red sides meet, and one corner where two blue sides meet. At each such intersection, there is a two-way door connecting the two diagonally opposite corners at which sides of different colours meet. After Merlin paints the walls, Morgana then places some knights in the labyrinth. The knights can walk through doors, but cannot walk through walls.

Let $k(\mathcal{L})$ be the largest number k such that, no matter how Merlin paints the labyrinth $\mathcal{L}$, Morgana can always place at least k knights such that no two of them can ever meet. For each n , what are all possible values for $k(\mathcal{L})$, where $\mathcal{L}$ is a labyrinth with n walls?

Problem 10.12 (ISL 2019). Let $n > 1$ be an integer. Suppose we are given $2n$ points in the plane such that no three of them are collinear. The points are to be labelled $A_1, A_2, \dots, A_{2n}$ in some order. We then consider the $2n$ angles

$$\angle A_1 A_2 A_3, \angle A_2 A_3 A_4, \dots, \angle A_{2n-2} A_{2n-1} A_{2n}, \angle A_{2n-1} A_{2n} A_1, \angle A_{2n} A_1 A_2$$

. We measure each angle in the way that gives the smallest positive value (i.e. between 0° and 180°). Prove that there exists an ordering of the given points such that the resulting $2n$ angles can be separated into two groups with the sum of one group of angles equal to the sum of the other group.

Problem 10.13 (LMAO 2021). Cat keeper Pranjal places cats $C_1, C_2, \dots, C_n$ on the plane. The cats wear magic helmets that can restrict their field of vision to an angle of their choosing. They now tilt their heads to restrict their vision so that no cat can see another cat. If the field of vision of C_i is α_i , then the score of the cats is defined to be $\sum_{i=1}^n \alpha_i$. Show that Pranjal can prevent the cats from scoring more than $1000 \ln n$, but that the cats can always score at least $\pi \ln n$.

Problem 10.14 (IMO 2020/6). Prove that there exists a positive constant c such that the following statement is true:

Consider an integer $n > 1$, and a set $\mathcal{S}$ of n points in the plane such that the distance between any two different points in $\mathcal{S}$ is at least 1. It follows that there is a line ℓ separating $\mathcal{S}$ such that the distance from any point of $\mathcal{S}$ to ℓ is at least $cn^{-1/3}$.

References/Resources

Here are some resources I find useful and would recommend along with some resources I referred to in the making of this handout.

- Yufei Zhao's handouts: Cyclic Quadrilaterals – The Big Picture, Circles, Three Lemmas in Geometry etc. All can be found at <https://yufeizhao.com/olympiad/>
- Some other Canadian IMO training handouts, can be found at: <https://sites.google.com/site/imocanada/home>
- Evan Chen's BMC Parallelograms handout
- Other Evan Chen handouts
- Ed Barbeau's Geometry Handout on transformations (can be found on Canada Winter 2008)
- Matthew Brennan's Combinatorial Geometry handout from Canada Winter 2019
- Titu's 107, 108 Geometry Problems books

Problem Sources to try:

- All Russian Math Olympiad
- Sharygin Math Olympiad
- Iranian Geometry Math Olympiad
- KoMaL problems
- Previous RMOs, INMOs
- British Math Olympiad Round 1, 2
- Canada Math Olympiad
- Eric Shen's Geometry at its best collection
- USA Ersatz Olympiad
- IMO Shortlist and IMO :)